MedTrack: AWS Cloud-Enabled Healthcare Management System

Project Description:

In today's fast-evolving healthcare landscape, efficient communication and coordination between doctors and patients are crucial. MedTrack is a cloud-based healthcare management system that streamlines patient doctor interactions by providing a centralized platform for booking appointments, managing medical histories, and enabling diagnosis submissions. To address these challenges, the project utilizes Flask for backend development, AWS EC2 for hosting, and DynamoDB for managing data. MedTrack allows patients to register, log in, book appointments, and submit diagnosis reports online. The system ensures real-time notifications, enhancing communication between doctors and patients regarding appointments and medical submissions. Additionally, AWS Identity and Access Management (IAM) is employed to ensure secure access control to AWS resources, allowing only authorized users to access sensitive data. This cloud-based solution improves accessibility and efficiency in healthcare services for all users.

Scenario 1: Efficient Appointment Booking System for Patients

In the MedTrack system, AWS EC2 provides a reliable infrastructure to manage multiple patients accessing the platform simultaneously. For example, a patient can log in, navigate to the appointment booking page, and easily submit a request for an appointment. Flask handles backend operations, efficiently retrieving and processing user data in real-time. The cloud-based architecture allows the platform to handle a high volume of appointment requests during peak periods, ensuring smooth operation without delays.

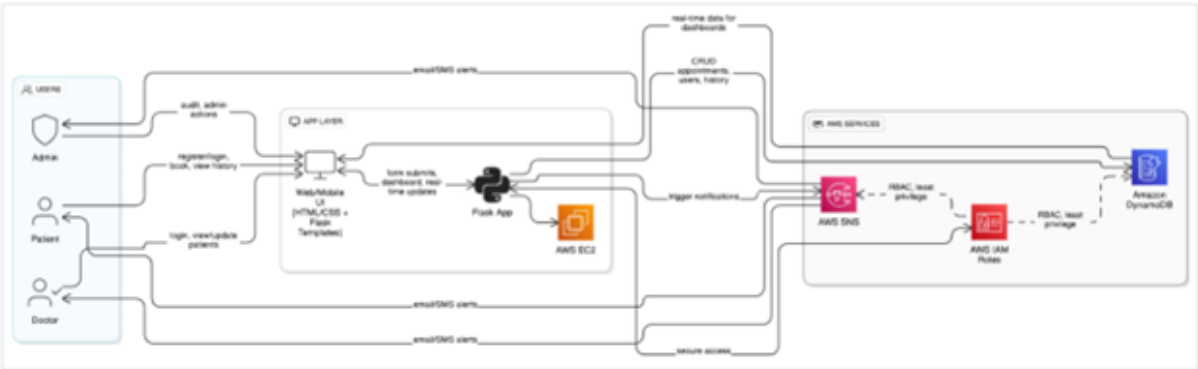
Scenario 2: Secure User Management with IAM

MedTrack utilizes AWS IAM to manage user permissions and ensure secure access to the system. For instance, when a new patient registers, an IAM user is created with specific roles and permissions to access only the features relevant to them. Doctors have their own IAM configurations, allowing them access to patient records and appointment details while maintaining strict security protocols. This setup ensures that sensitive data is accessible only to authorized users.

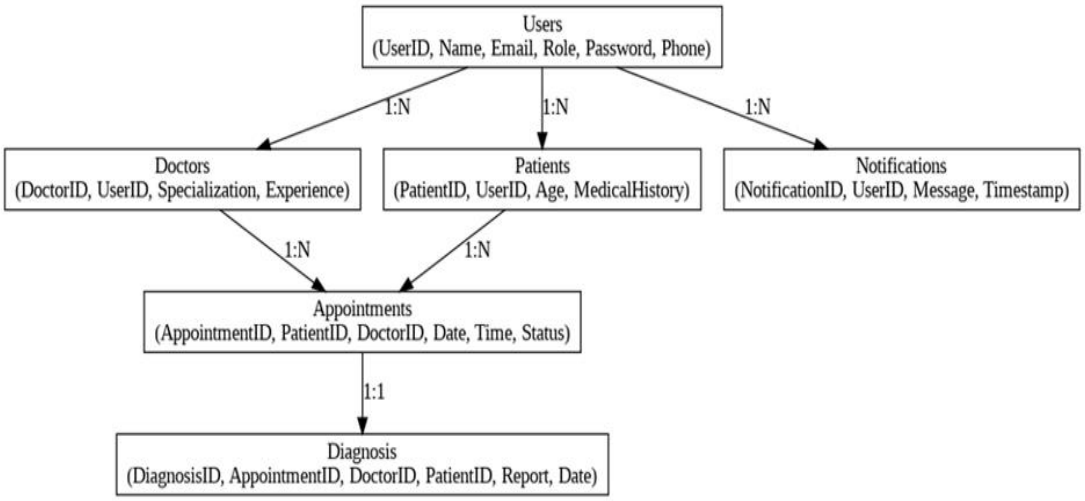
Scenario 3: Easy Access to Medical History and Resources

The MedTrack system provides doctors and patients with easy access to medical histories and relevant resources. For example, a doctor logs in to view a patient's medical history and upcoming appointments. They can quickly access, and update records as needed. Flask manages real-time data fetching from DynamoDB, while EC2 hosting ensures the platform performs seamlessly even when multiple users access it simultaneously, offering a smooth and uninterrupted user experience.

AWS Architecture



Entity Relationship (ER)Diagram:



Pre-requisites:

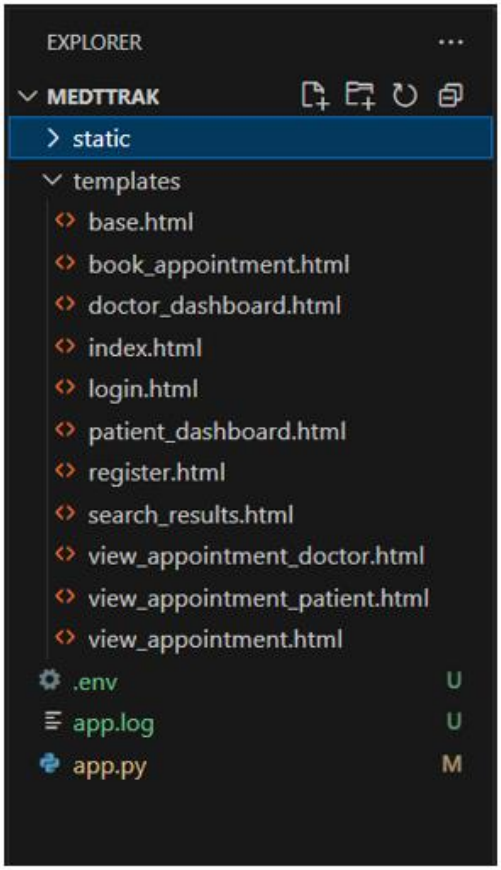
1. AWS Account Setup
2. Understanding IAM
3. Amazon EC2 Basics
4. DynamoDB Basics
5. SNS Overview
6. Git Version Control

Project Workflow

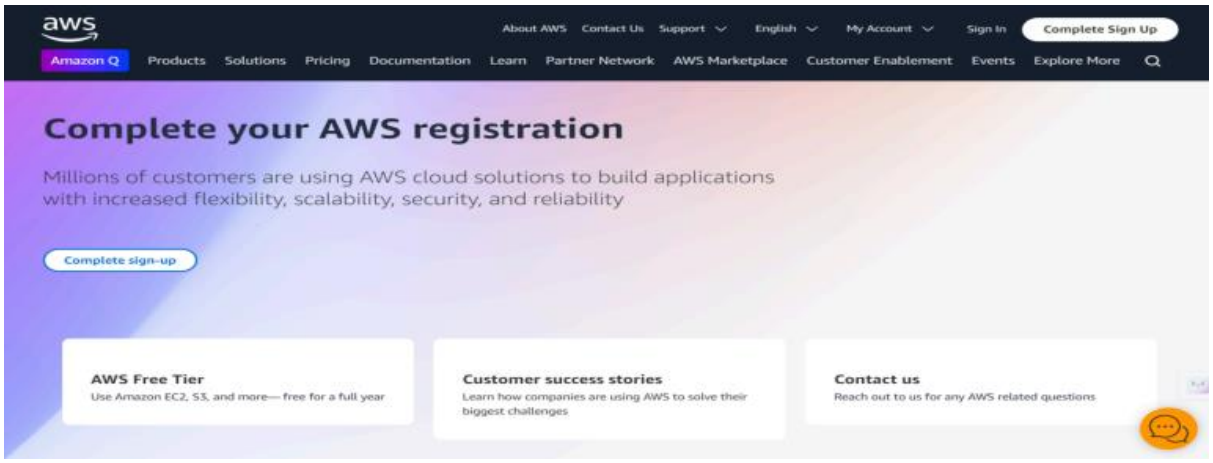
1. **Backend Development and Application Setup**
   - Develop the Backend Using Flask.
   - Integrate AWS Services Using boto3
2. **AWS Account Setup and Login**
   - Set up an AWS account if not already done
   - Log in to the AWS Management Console
3. **DynamoDB Database Creation and Setup**
   - Create a DynamoDB Table
   - Configure Attributes for User Data and Book Requests
4. **SNS Notification Setup**
   - Create SNS topics for book request notifications
   - Subscribe users and library staff to SNS email notifications
5. **IAM Role Setup**
   - Create IAM Role
   - Attach Policies
6. **EC2 Instance Setup**
   - Launch an EC2 instance to host the Flask application
   - Configure security groups for HTTP, and SSH access
7. **Deployment on EC2**
   - Upload Flask Files
   - Run the Flask App
8. **Testing and Deployment**

- Conduct functional testing to verify user registration, login, book requests, and notifications

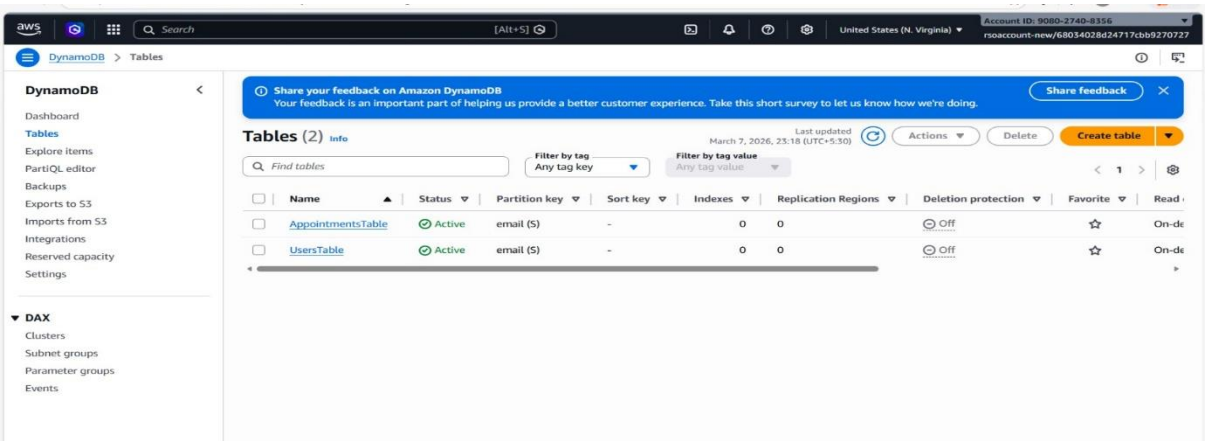
Milestone 1: Backend Development and Application Setup



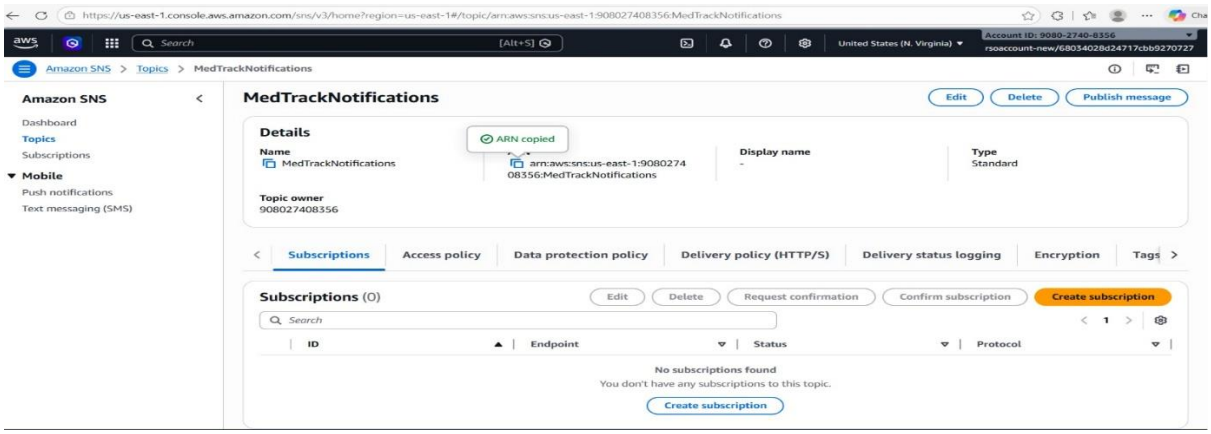
Milestone 2: AWS Account Setup and Login



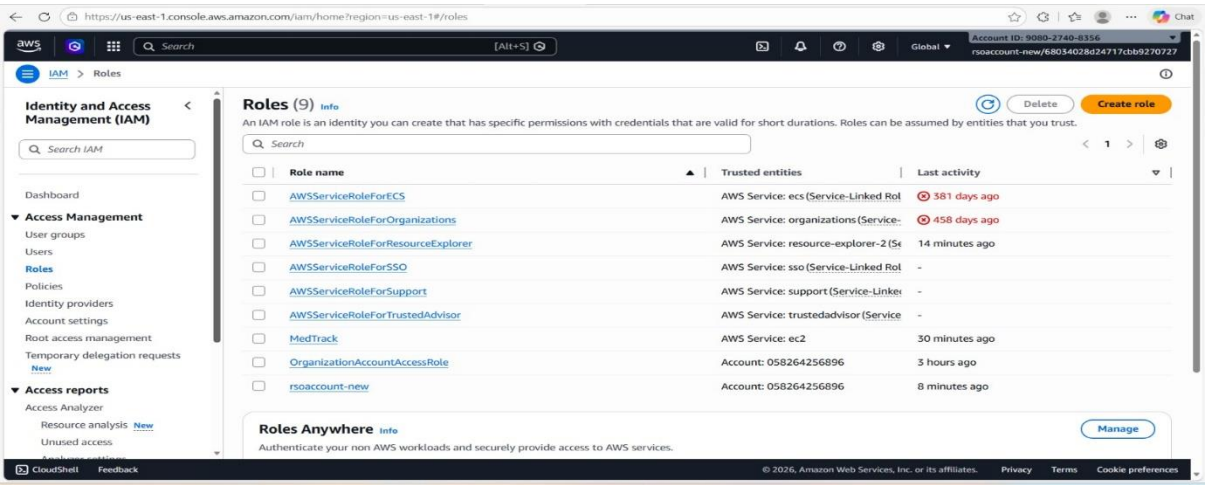
Milestone 3: DynamoDB Database Creation and Setup



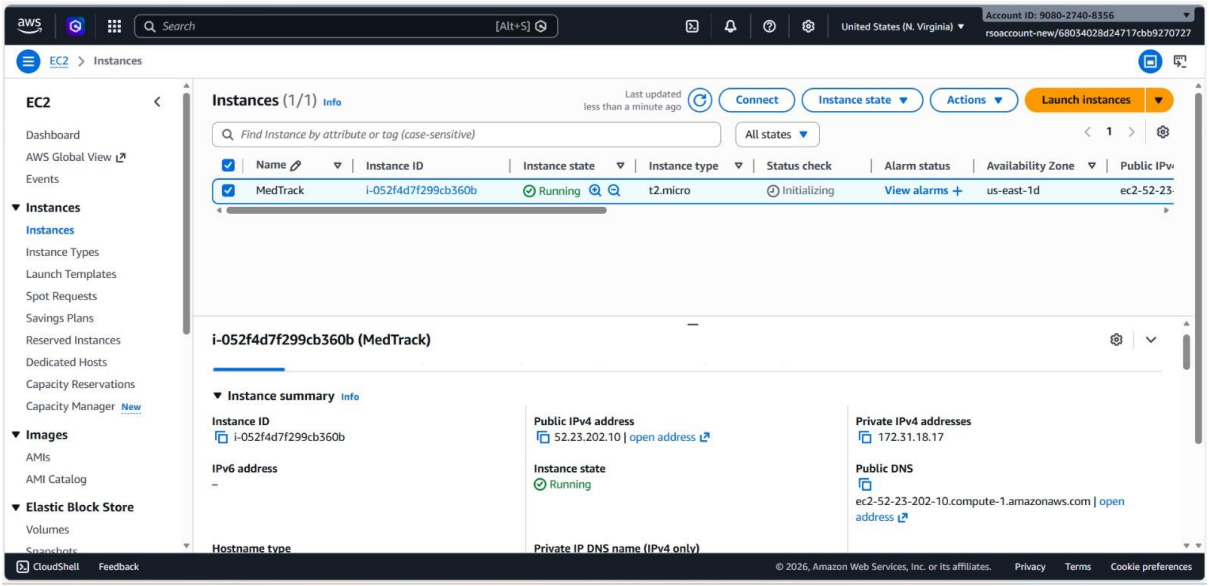
Milestone 4: SNS Notification Setup



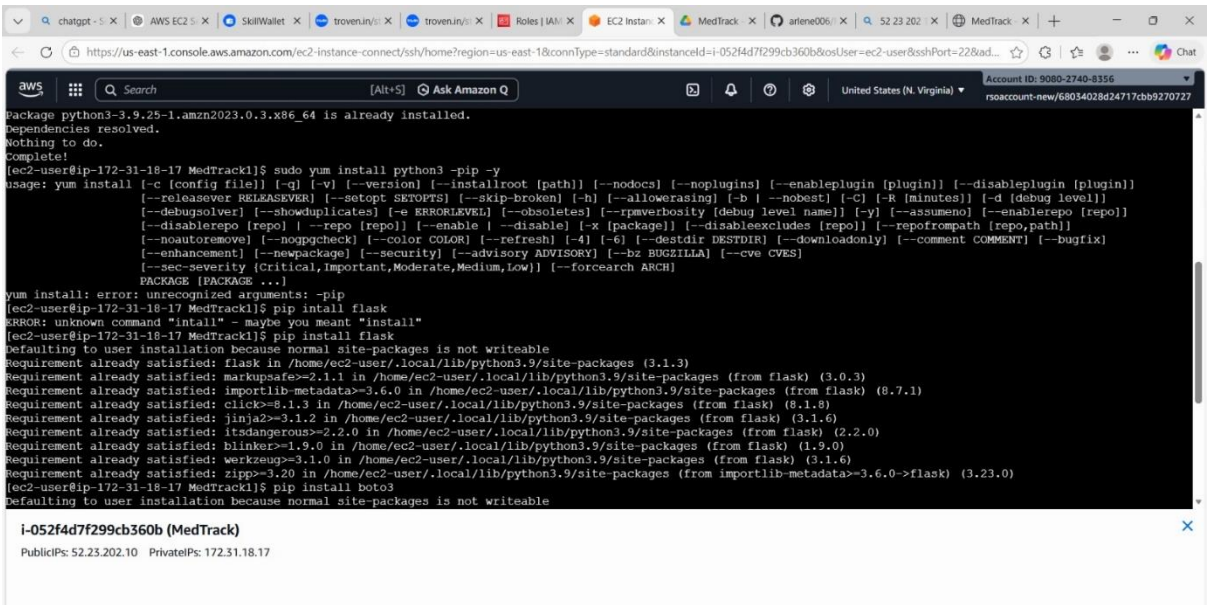
Milestone 5: IAM Role Setup



Milestone 6: EC2 Instance Setup

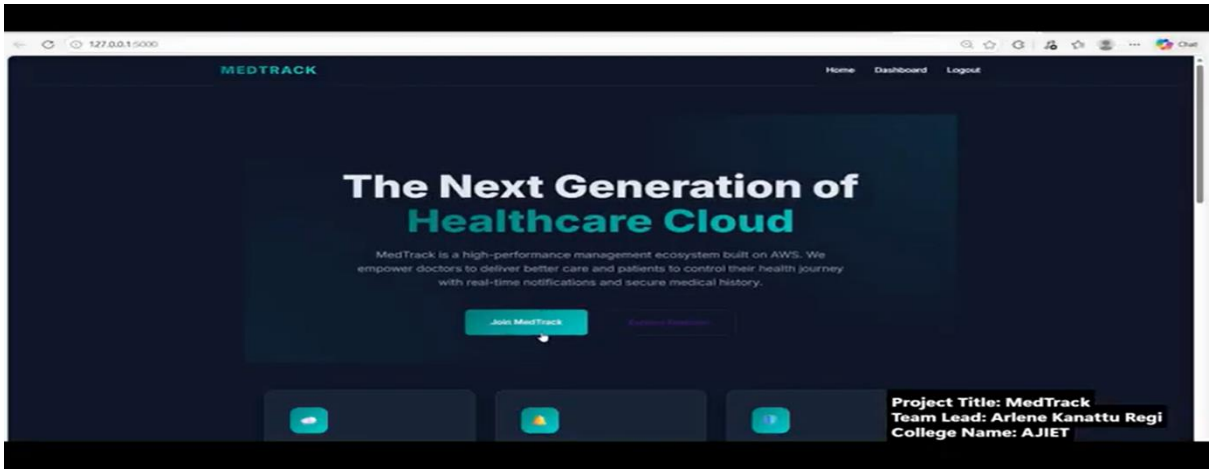


Milestone 7: Deployment on EC2

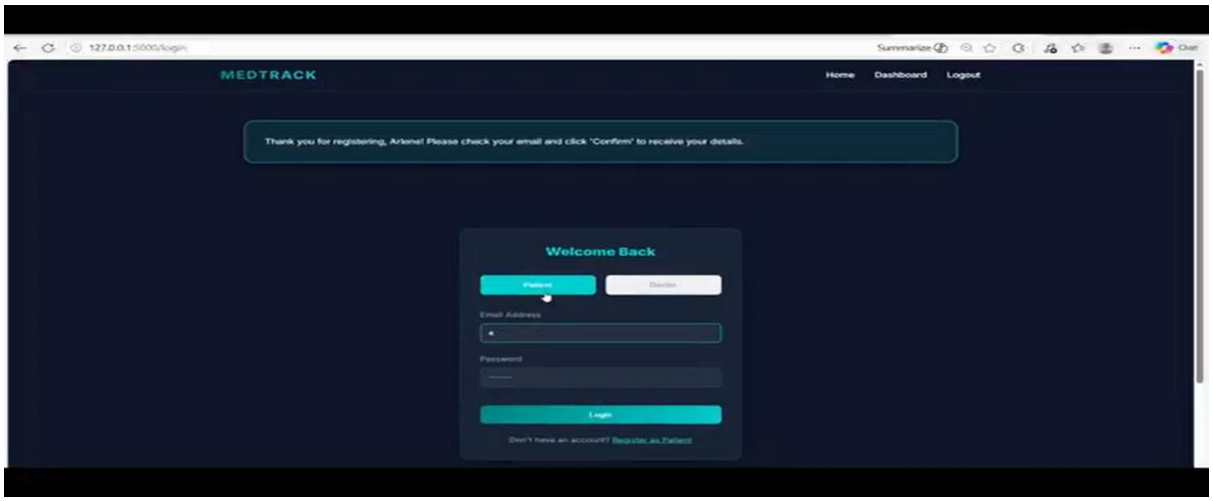


Milestone 8: Testing and Deployment

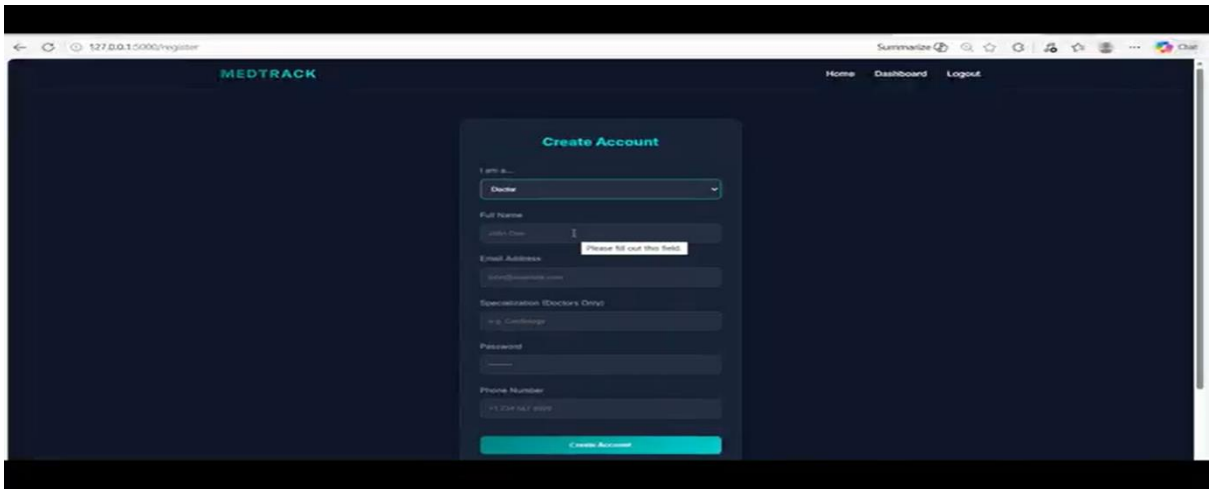
Home Page:



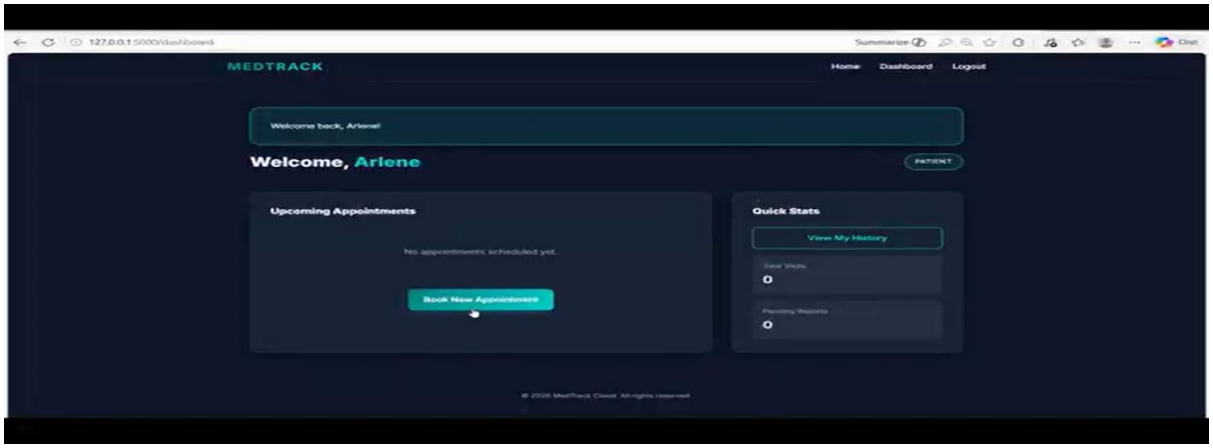
Login Page:



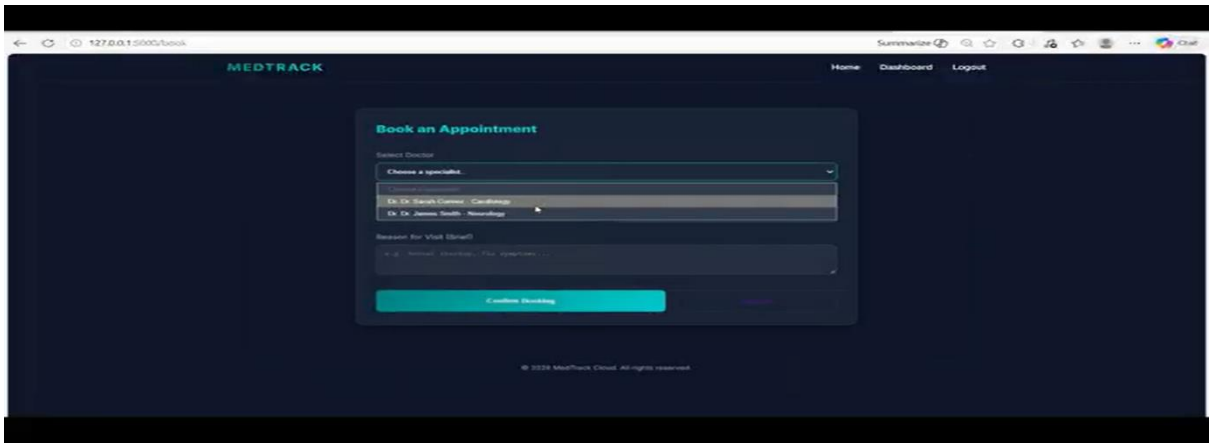
Register Page:



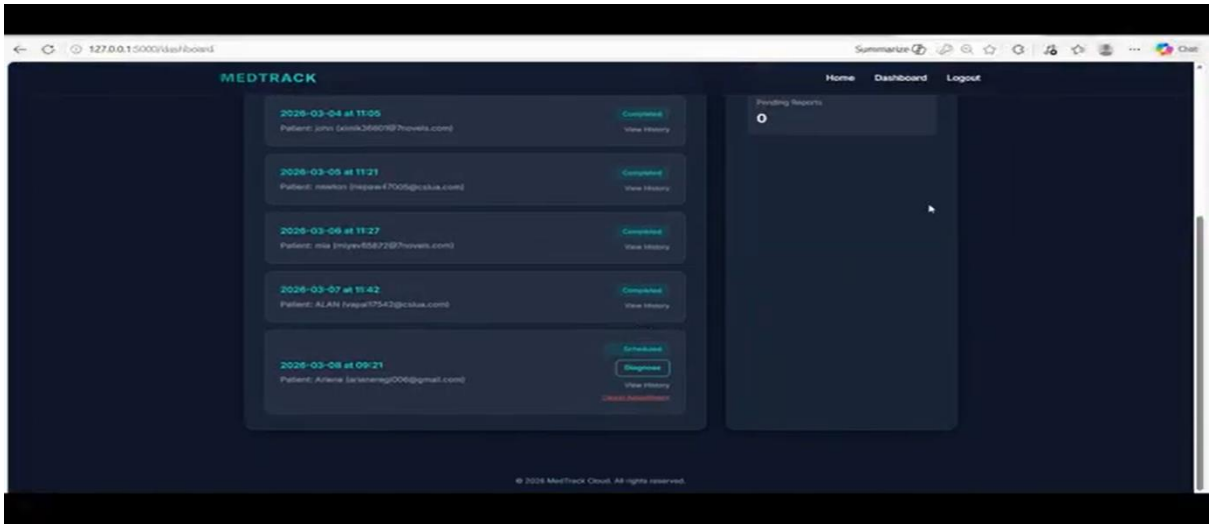
Patient Dashboard:



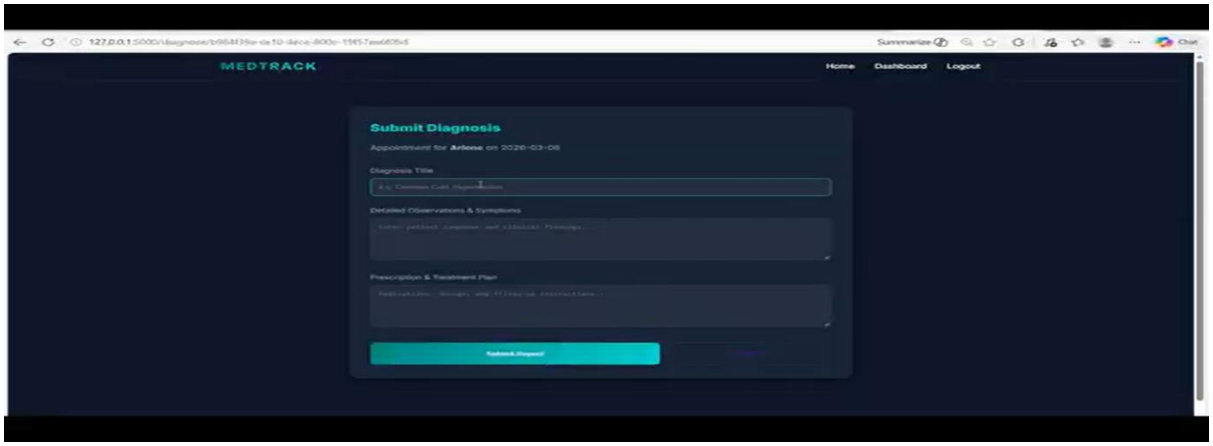
Book Appointment Page:



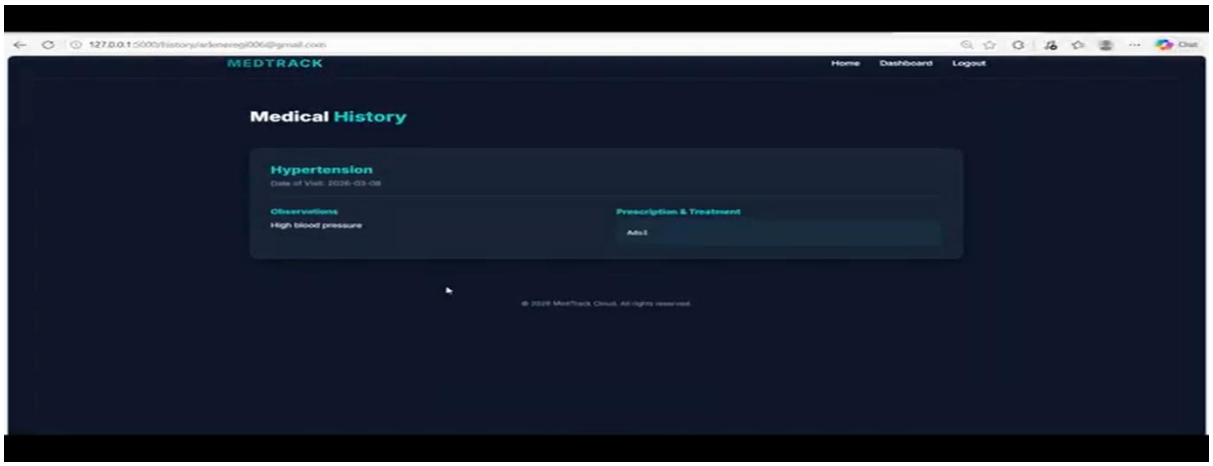
Doctor Dashboard:



Submit Diagnosis Page:

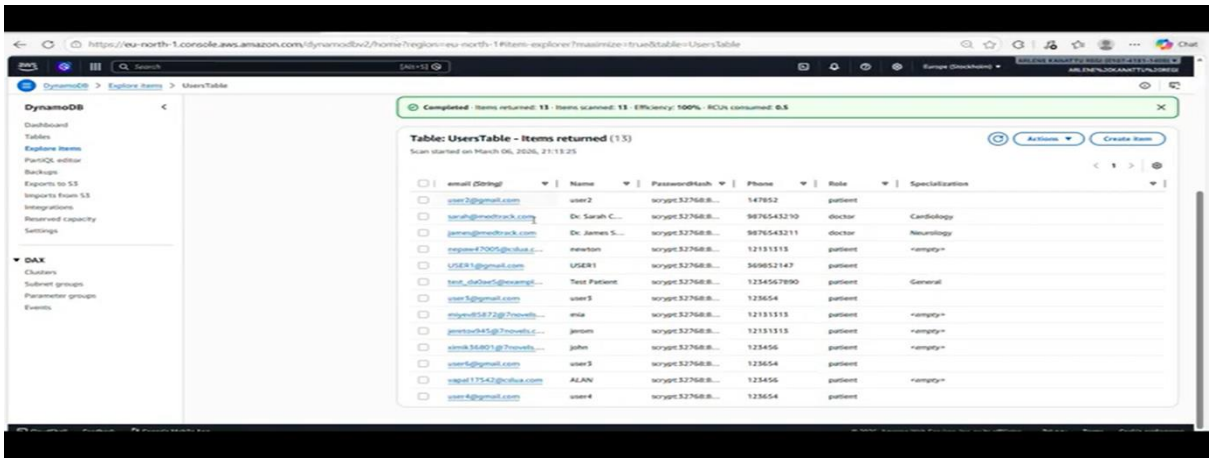


Medical History Page:

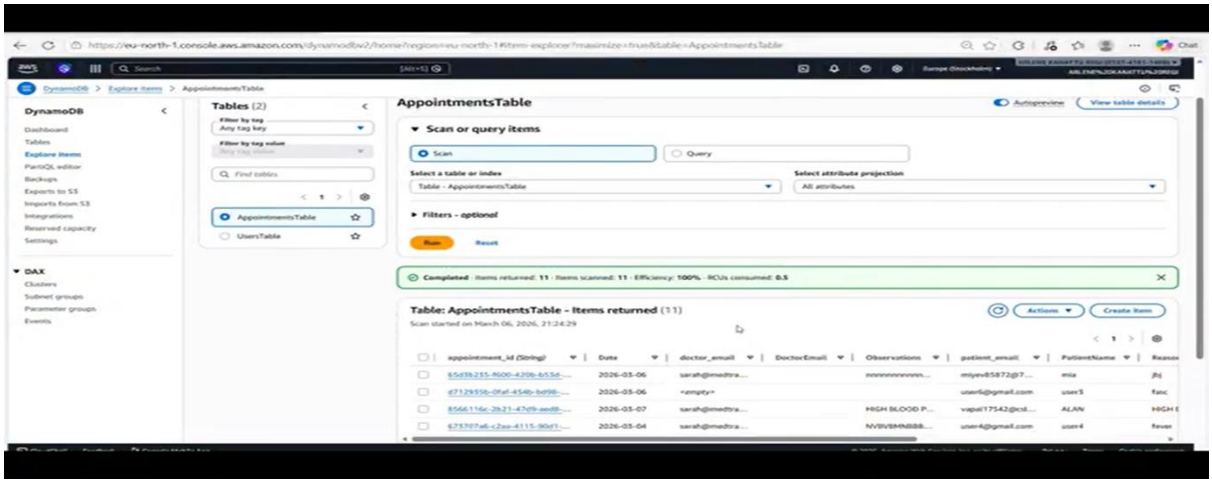


Dynamo DB Database Updations:

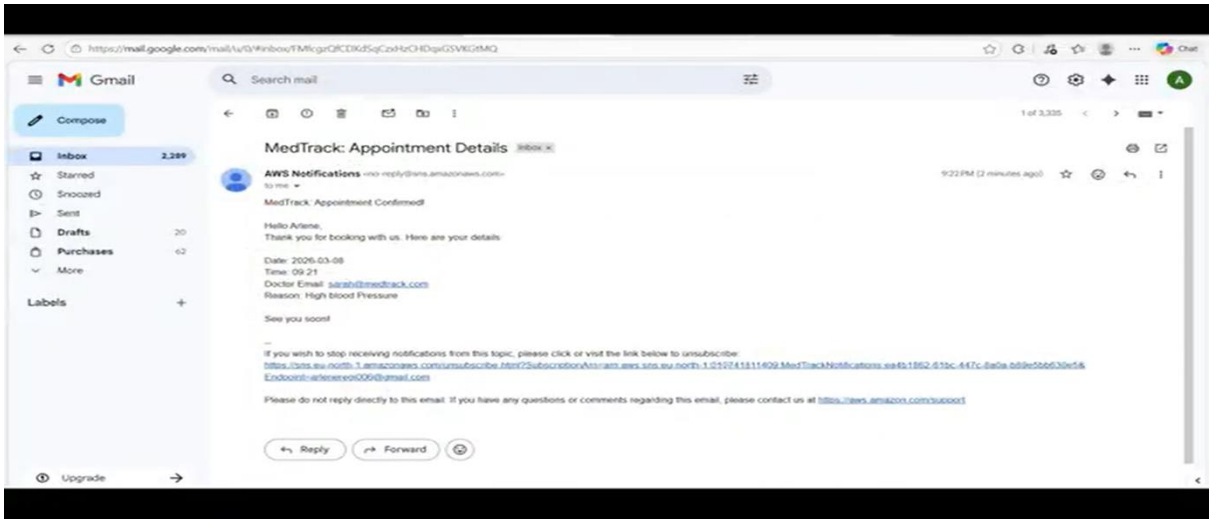
1. **Users table:**



2. **Appointments Table:**



3. **Mail Sent To User:**



Conclusion

The MedTrack application has been successfully developed and deployed using a robust cloud based architecture tailored for modern healthcare environments. Leveraging AWS services such as EC2 for hosting, DynamoDB for secure and scalable patient data management, and SNS for real-time alerts, the platform ensures reliable and efficient access to essential medical tracking services. This system addresses critical challenges in healthcare such as managing patient records, monitoring medication schedules, and ensuring timely communication between healthcare providers and patients.

The cloud-native approach enables seamless scalability, allowing MedTrack to support increasing numbers of users and data without compromising performance or reliability. The integration of Flask with AWS ensures smooth backend operations, including patient registration, medication reminders, and health updates. Thorough testing has validated that all features—from user onboarding to alert notifications—function reliably and securely.

In conclusion, the MedTrack application delivers a smart, efficient solution for modernizing healthcare management, improving patient care, and streamlining communication between medical staff and patients. This project highlights the transformative power of cloud-based technologies in solving real-world challenges in the healthcare sector.